



Genomic epidemiology of multidrug-resistant clinical *Acinetobacter baumannii* in Bangladesh

Md Abu Sayem Khan, Sudipta Chowdhury Chaity, Md Arman Hosen, Sabita Rezwana Rahman*

Department of Microbiology, University of Dhaka, Dhaka, Bangladesh

ARTICLE INFO

Keywords:

A. baumannii

Genome sequence

Bangladesh

Molecular epidemiology

AMR

ABSTRACT

The rising frequency of multidrug-resistant (MDR) *Acinetobacter baumannii* infections represents a significant public health challenge in Bangladesh. Genomic analysis of bacterial pathogens enhances surveillance and control efforts by providing insights into genetic diversity, antimicrobial resistance (AMR) profiles, and transmission dynamics. In this study, we conducted a comprehensive bioinformatic analysis of 82 whole-genome sequences (WGS) of *A. baumannii* from Bangladesh to understand their genomic epidemiological characteristics. WGS of the MDR and biofilm-forming *A. baumannii* strain S1C revealed the presence of 28 AMR genes, predicting its pathogenicity and classification within sequence type ST2. Multi-locus sequence typing (MLST) genotyping suggested heterogeneity in the distribution of clinical *A. baumannii* strains in Bangladesh, with a predominance of ST575. The resistome diversity was evident from the detection of 82 different AMR genes, with antibiotic inactivation being the most prevalent resistance mechanism. All strains were predicted to be multidrug-resistant. The observed virulence genes were associated with immune evasion, biofilm formation, adherence, nutrient acquisition, effector delivery, and other mechanisms. Mobile genetic elements carrying AMR genes were predicted in 68.29% ($N = 56$) of the genomes. The “open” state of the pan-genome and a high proportion of accessory genes highlighted the genome plasticity and diversity of *A. baumannii* in Bangladesh. Additionally, phylogenomic analysis indicated clustering of *A. baumannii* strains into three separate clades according to sequence type. In summary, our findings offer detailed insights into the genomic landscape of *A. baumannii* in Bangladesh, contributing to our understanding of its epidemiology and pathogenicity and informing strategies to combat this pathogen.

1. Introduction

Acinetobacter baumannii has emerged as a leading cause of hospital-acquired or nosocomial infections, including pneumonia, meningitis, bloodstream infections, and urinary tract infections, leading to considerable morbidity and mortality among immunocompromised patients (Cui et al., 2023). *A. baumannii* exhibits a high degree of resistance to existing antimicrobial classes, particularly carbapenems and colistin, which are considered last-resort treatments (Wareth et al., 2021). Therefore, the widespread dissemination of multidrug-resistant (MDR) *A. baumannii* (MDRAB) strains poses a substantial public health threat globally. A meta-analysis determined the worldwide prevalence of MDRAB in hospital-acquired pneumonia or ventilator-associated pneumonia patients is approximately 79.9%, with mortality rates ranging from 37.2% to 48.1% (Lim et al., 2019). According to the Centers for Disease Control and Prevention (CDC), MDRAB causes up to 20% of

infections in critical care units, constituting the majority of clinically relevant strains (Nasr, 2020). Considering the severity of infections, level of resistance, and limited treatment options, the World Health Organization (WHO) listed carbapenem-resistant *A. baumannii* (CRAB) as the top “critical” priority pathogen to enhance global preparedness and accelerate antibiotic discovery (Tacconelli et al., 2017).

The success of *A. baumannii* as an emerging nosocomial pathogen over the last two decades might be attributed to adaptation to hospital environments, development of MDR, expression of multiple virulence factors, and the propensity for gene transmission through a diverse mobilome (Brown and Wright, 2016; Kumkar et al., 2022). According to a study on *A. baumannii* colonization ability on inanimate surfaces, clinical isolates exhibited greater capacity to form biofilms on abiotic surfaces compared to non-clinical isolates. Therefore, this robust biofilm-forming capacity is considered a crucial contributing factor promoting chronic and persistent infections in hospital settings (Jacobs

* Corresponding author.

E-mail address: sabita_rahman@du.ac.bd (S.R. Rahman).

<https://doi.org/10.1016/j.meegid.2024.105656>

Received 14 May 2024; Received in revised form 4 July 2024; Accepted 5 August 2024

Available online 6 August 2024

1567-1348/© 2024 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

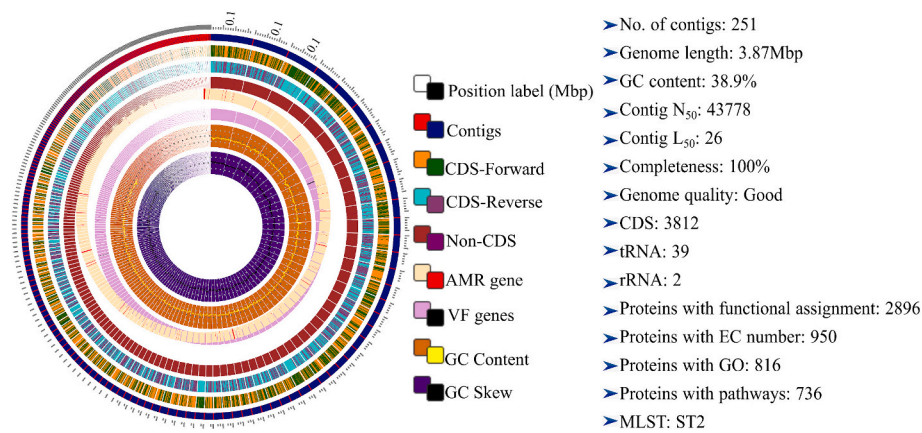


Fig. 1. The circular genome map of *A. baumannii* with the characteristics of draft genome.

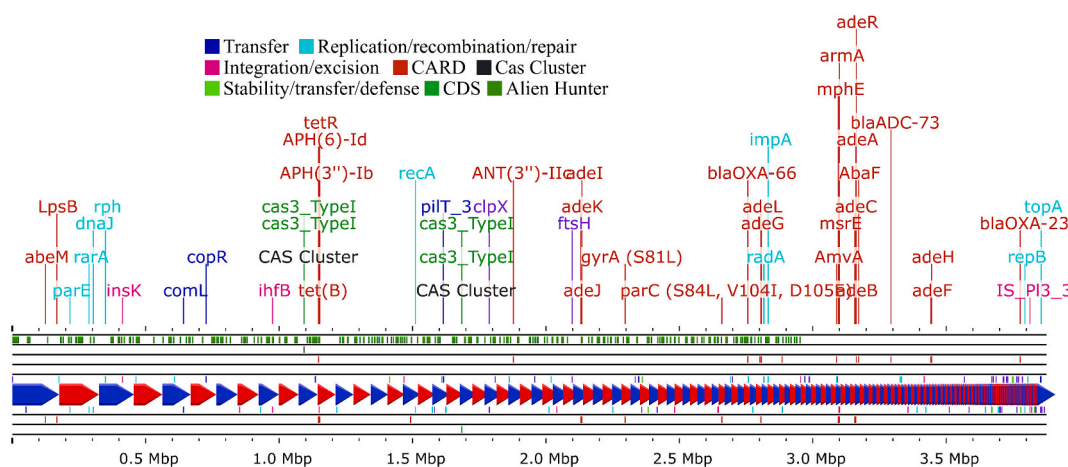


Fig. 2. Genome annotation of *A. baumannii* S1C.

et al., 2014; Roy et al., 2022). In addition to biofilm formation, *A. baumannii* possesses a notable ability to acquire and transfer AMR genes, facilitating the global emergence and transmission of MDRAB (Gao et al., 2024). Among diverse resistance mechanisms, carbapenemase production, particularly class D β -lactamases (oxacillinases), is one of the most common mechanisms for carbapenem resistance in CRAB. Oxacillinases such as OXA-23, OXA-24, and OXA-58 are frequently found among *A. baumannii* strains. These enzymes are encoded by various alleles with differing hydrolytic capacities (Odih et al., 2023). Dissemination and acquisition of carbapenemases are frequently associated with plasmids and other mobile genetic elements (MGEs), as evident from previous studies (Baleivannualala et al., 2023). For example, Nigro et al. reported that the OXA-23 gene is usually flanked by an insertion sequence (IS), ISAbal, which provides the strong promoter required for clinically significant resistance levels. The study also described the significance of Tn6022, one of the four ISAbal-associated transposons, in the global dissemination of OXA-23 in *A. baumannii* (Nigro and Hall, 2016).

Studies on the phenotypic and genotypic investigation of AMR in clinical isolates of *A. baumannii* in Bangladesh have reported the extent of resistance as well as the prevalence and diversity in AMR-associated genes. Most studies used disc diffusion methods along with gene-specific polymerase chain reaction (PCR) to depict the AMR scenario in clinical *A. baumannii*. In one study, all clinical isolates were ciprofloxacin-resistant, with 91.8% exhibiting resistance to imipenem, meropenem, and gentamicin, and 59% classified as MDR (Farzana et al., 2022). Islam et al. reported similar carbapenem resistance trends, with 100% and 99% resistance to meropenem and imipenem, respectively

(Islam et al., 2023). The high prevalence of metallo- β -lactamase (MBL) genes was also evident. A study on MBL gene molecular profiling determined a high prevalence of *bla*NDM-1 (83.33%), *bla*VIM (66.67%), and *bla*IMP (41.67%) in *A. baumannii* (Uddin et al., 2020). Similarly, Khatun et al. reported the detection of *bla*NDM-1 in 92% of isolates (Khatun et al., 2016). Later, genomic sequencing provided more genomic details into *A. baumannii* drug resistance and virulence. For example, whole-genome sequencing revealed the co-existence of four different β -lactamase classes (*bla*PER-7, *bla*TEM-116: Class A, *bla*NDM-1: Class B, *bla*ADC-26: Class C, and *bla*OXA-23, *bla*OXA-51: Class D) in the *A. baumannii* strain DMC-32a from a wound swab (Jahan et al., 2021). Apart from individual genome analysis, a comparative genomic analysis elucidated the resistome, mobilome, and virulome structures of 14 extensive drug-resistant (XDR) *A. baumannii* belonging to different clonal lineages (Rahman et al., 2022). A comparison of this evidence suggested that the *A. baumannii*'s resistance repertoire is evolving with the acquisition of new AMR genes, highlighting the importance of continuous monitoring for emerging clonal types and resistance genes.

Understanding the mechanisms of acquired AMR and virulence-associated genes, as well as genomic diversity among MDRAB isolates, would provide valuable information on their dissemination patterns within regions. Whole-genome sequencing (WGS) allows for a detailed examination of the genomic composition of *A. baumannii*, beyond traditional phenotypic and genotypic characterization (Kumkar et al., 2022). While previous studies have separately analyzed and reported the genetic makeup of *A. baumannii* in Bangladesh, their combined analysis remains unexplored, representing a significant knowledge gap in the genomic landscape of this pathogen. In this study, we aimed to

Table 1

Acquired antimicrobial resistance genes and virulence factors of *A. baumannii* S1C.

Gene		Product	Resistance/ Function
Antimicrobial resistance genes	<i>bla_{OXA-66}</i>	OXA-51 family carbapenem-hydrolyzing class D beta-lactamase	Carbapenem
	<i>bla_{OXA-23}</i>	carbapenem-hydrolyzing class D beta-lactamase OXA-23	
	<i>mph(E)</i>	Mph(E) family macrolide 2'-phosphotransferase	Macrolide
	<i>msr(E)</i>	ABC-F type ribosomal protection protein Msr(E)	
	<i>bla_{ADC-73}</i>	class C extended-spectrum beta-lactamase ADC-73	Cephalosporin
	<i>tet(B)</i>	tetracycline efflux MFS transporter Tet(B)	Tetracycline
	<i>aph(6)-Id</i>	aminoglycoside O-phosphotransferase APH(6)-Id	Aminoglycoside
	<i>aph(3'')-Ib</i>	aminoglycoside O-phosphotransferase APH(3'')-Ib	
	<i>armA</i>	ArmA family 16S rRNA (guanine(1405)-N(7))-methyltransferase	
Virulence genes	<i>ant(3'')-IIa</i>	aminoglycoside nucleotidyltransferase ANT(3'')-IIa	
	<i>pilB</i> , <i>pilU</i> <i>pilM</i> <i>pilJ</i>	Type IV-A pilus assembly ATPase Pilus assembly protein Methyl-accepting chemotaxis protein	Adherence
	<i>bfmS</i>	Signal transduction histidine kinase	Regulation
	<i>pgaA</i>	poly-beta-1,6 N-acetyl-D-glucosamine export porin	Biofilm
	<i>csuD</i>	Csu pilus usher protein	
	<i>wza</i> , <i>wzb</i> , <i>wzc</i> <i>lpxD</i>	EPS synthesis and export	
		UDP-3-O-(3-hydroxymyristoyl) glucosamine N-acyltransferase	Immune modulation
	<i>basD</i>	Acinetobactin biosynthesis protein BasD	Nutritional/ Metabolic factor
	<i>barA</i>	Siderophore efflux system	
	<i>tssM</i>	Type VI secretion system membrane subunit	Effector delivery system
	<i>vgrG</i> / <i>tssI</i> <i>gspK</i> , <i>gspD</i>	Type VI secretion system tip protein general secretion pathway protein K	

comprehensively characterize the resistome, virulome, and mobilome of *A. baumannii* using whole-genome analysis. We conducted WGS analysis of one *A. baumannii* strain and retrieved additional whole-genome sequences from databases, representing *A. baumannii* strains isolated from clinical patients in Bangladesh. By elucidating the genetic determinants associated with AMR and virulence, their composition and distribution, as well as horizontal gene transfer mechanisms, we aim to provide valuable insights into the genomic epidemiology and evolution of clinical *A. baumannii* strains in Bangladesh.

2. Materials and methods

2.1. Isolation and presumptive identification of *Acinetobacter baumannii*

A. baumannii S1C was isolated from a tracheal aspirate sample of a 70-year-old man admitted to the intensive care unit (ICU) of a tertiary

care hospital in Dhaka, Bangladesh in 2023. The isolate was grown on MacConkey agar (Oxoid, UK), and a panel of biochemical tests (IMViC, oxidase, catalase, Kligler iron agar, and hemolysis) were performed for presumptive identification.

2.2. Antimicrobial susceptibility testing

Disc diffusion method was followed for the characterization of the susceptibility of S1C to 13 clinically important antibiotics: amikacin (30 µg), ceftriaxone (30 µg), cefotaxime (30 µg), cefepime (30 µg), meropenem (10 µg), imipenem (5 µg), ciprofloxacin (5 µg), piperacillin+tazobactam (100/10 µg), tigecycline (15 µg), linezolid (30 µg), aztreonam (30 µg), co-trimoxazole (25 µg) and gentamicin (10 µg) (Oxoid, UK). The turbidity of the freshly grown culture of the isolate was standardized to 0.5 McFarland. Then, a lawn of bacteria was prepared over the Mueller Hinton agar (Oxoid, UK) by homogenously spreading the culture and the discs were placed on the surface of the agar. Following incubation at 37 °C for 18-20 h, the diameter of the zone of inhibition was measured. Bacterial susceptibility to antibiotics was determined according to the Clinical and Laboratory Standard Institute (CLSI) guidelines (2023).

2.3. Biofilm formation assay

The *in vitro* biofilm formation ability of strain S1C was analyzed using crystal violet microtiter plate assay. Briefly, the freshly grown culture of S1C was inoculated in a microtiter plate (Corning, USA) and incubated at 37 °C for 24 h to allow biofilm formation. Then, the non-adherent cells were washed with normal saline and the wells were stained with 200 µl of 0.1% crystal violet (Sigma-Aldrich, USA) at room temperature for 30 min. The stained wells were washed again with normal saline and the remaining stain was solubilized using ethanol. The absorbance of the dissolved stain was measured at 492 nm. The extent of the biofilm formation was interpreted according to the criteria as described previously (Stepanović et al., 2007).

2.4. Whole genome sequencing and analysis

The genomic DNA of S1C was extracted using Wizard® Genomic DNA Purification Kit according to manufacturer's instructions. WGS was performed on Illumina NovaSeq 6000 via paired end sequencing (2 × 151 bp). The quality of the raw reads was assessed using FastQC (<http://www.bioinformatics.babraham.ac.uk/projects/fastqc/>). Then, Trimmomatic (Bolger et al., 2014) was used to remove adapter and then, the quality filtered reads were assembled using SPAdes-V 3.15.5 (Bankevich et al., 2012). The assembly characteristics and completeness of the draft genome were estimated using QUAST (Gurevich et al., 2013) and BUSCO (Simão et al., 2015), respectively. The Type Strain Genome Server (TYGS) (Meier-Kolthoff and Göker, 2019) was employed for whole genome phylogeny-based identification of S1C. The assembled sequence was uploaded in SpeciesFinder v2.0 (Larsen et al., 2014) to reconfirm *A. baumannii* species. The pathogenicity of S1C towards human hosts was predicted using PathogenFinder (Cosentino et al., 2013).

2.5. Retrieval of *Acinetobacter baumannii* sequences from databases

The whole genome sequences of the *A. baumannii* were retrieved from NCBI (<https://www.ncbi.nlm.nih.gov/>), BV-BRC (<https://www.bv-brc.org/>), and PubMLST (<https://pubmlst.org/>) (accessed in October 2023) along with available metadata. These databases were searched to download the raw sequences of confirmed clinical *A. baumannii* isolates reported from Bangladesh, in order to conduct a comprehensive genomic investigation on their virulence, antimicrobial resistance, and mobile genetic elements. For the NCBI database, the search parameters included the organism name '*Acinetobacter baumannii*', the country

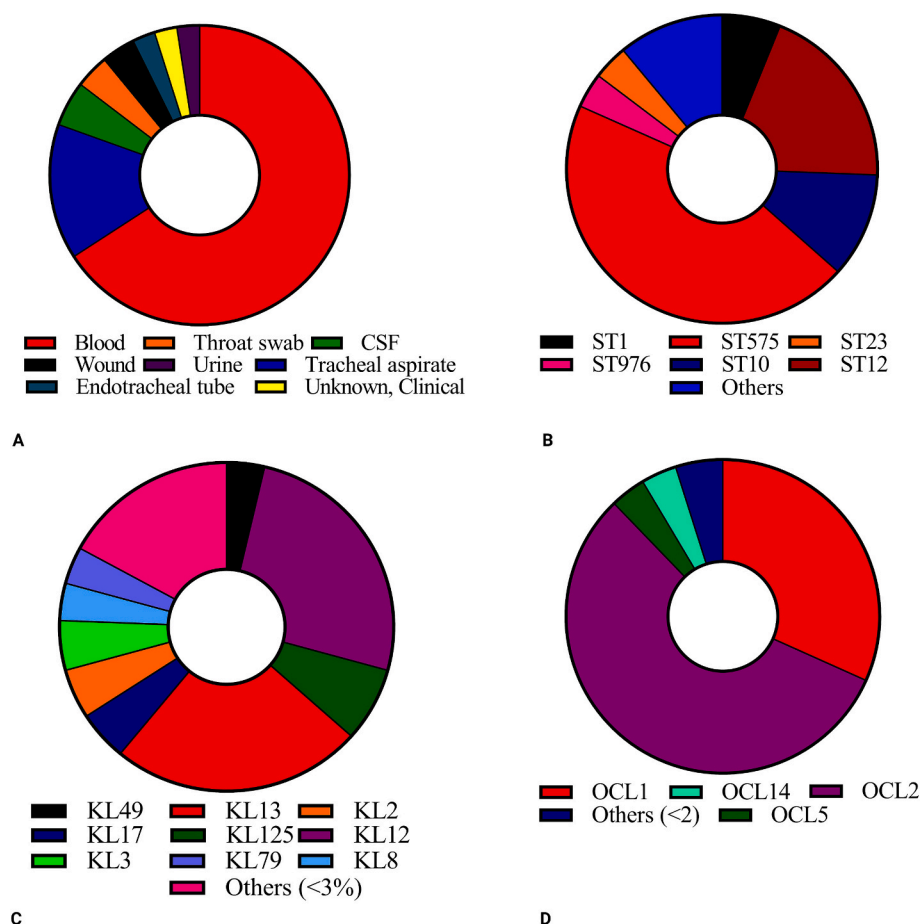


Fig. 3. Distribution of *A. baumannii* population in Bangladesh based on sample types (A), sequence types (B), capsular polysaccharide (C) and lipooligosaccharide (D).

'Bangladesh', and the collection date range '2015–2023'. Similar parameters were applied for searches in the BV-BRC and PubMLST databases. The selection criteria for the isolates included: (1) confirmation of the species as *A. baumannii*, (2) availability of metadata indicating the clinical origin of the isolates, (3) documentation of the isolation location as Bangladesh, and (4) the time period of isolation being between 2015 and 2023. In cases where a genome was found in multiple databases, duplicates were identified and removed to ensure each genome was included only once. The raw sequences were downloaded and processed using the same tools that were used to assemble the S1C genome, as described earlier. In cases where raw files were unavailable, the assembled genomes in FASTA format were retrieved. The details of analyzed sequences are listed in Supplementary table 1.

2.6. Quality assessment and annotation of assembled sequences

The assembly characteristics including the genome length, GC content, number of contigs and others were assessed using QUAST. Both RAST (Aziz et al., 2008) and PROKKA (Seemann, 2014) were used to annotate all assembled sequences. For multilocus sequence typing (MLST), we searched the sequence types of genomes against PubMLST (Jolley et al., 2018). In addition, the capsular polysaccharide (K locus) and lipooligosaccharide (OC locus) of *A. baumannii* were predicted by Kaptive 2.0 (Lam et al., 2022).

2.7. In silico AMR and virulence profiling

The AMR genes were identified using ABRicate (<https://github.com/tseemann/abricate>) against the Comprehensive Antimicrobial

Resistance Database (CARD) (Alcock et al., 2020) and ResFinder (Bortolaia et al., 2020) database (minimum identity = 80% and minimum coverage = 80%). The draft genomes were scanned against VFDB (Chen et al., 2005) database to identify virulence genes.

2.8. Characterization of mobile genetic elements

The integrated phage regions in the genomes were identified using PHASTER (Arndt et al., 2016). Plasmid sequences were identified and typed using ABRicate against PlasmidFinder database (Carattoli and Hasman, 2020). The studied genomes were subjected to VRprofile2 (Wang et al., 2022) to predict mobilome and MGE associated AMR genes. We also analyzed these sequences using BacAnt (Hua et al., 2021) and ISEScan (Xie and Tang, 2017) at default parameters to get more insights into the integrons and insertion sequences in the draft genomes.

2.9. Pan-genome analysis and construction of phylogenetic tree

We used Roary (Page et al., 2015) to perform the pan-genome analysis of 82 *A. baumannii* genomes including our strain S1C using the gff3 files generated by Prokka at 95% minimum blastp identity and 99% core definition threshold. The pan-genome matrix was visualized in Phandango (<https://jameshadfield.github.io/phandango/#/>). Moreover, the graphs showing the changes in pan genome were created using roary_plots script (https://github.com/sanger-pathogens/Roary/blob/master/bin/create_pan_genome_plots.R). The core genome alignment file from Roary was subjected to SNP-sites (Page et al., 2016) to call SNPs in core genome. Then, a maximum-likelihood phylogenetic tree was built using IQ-TREE 2.0 (Minh et al., 2020) with 1000 bootstrap

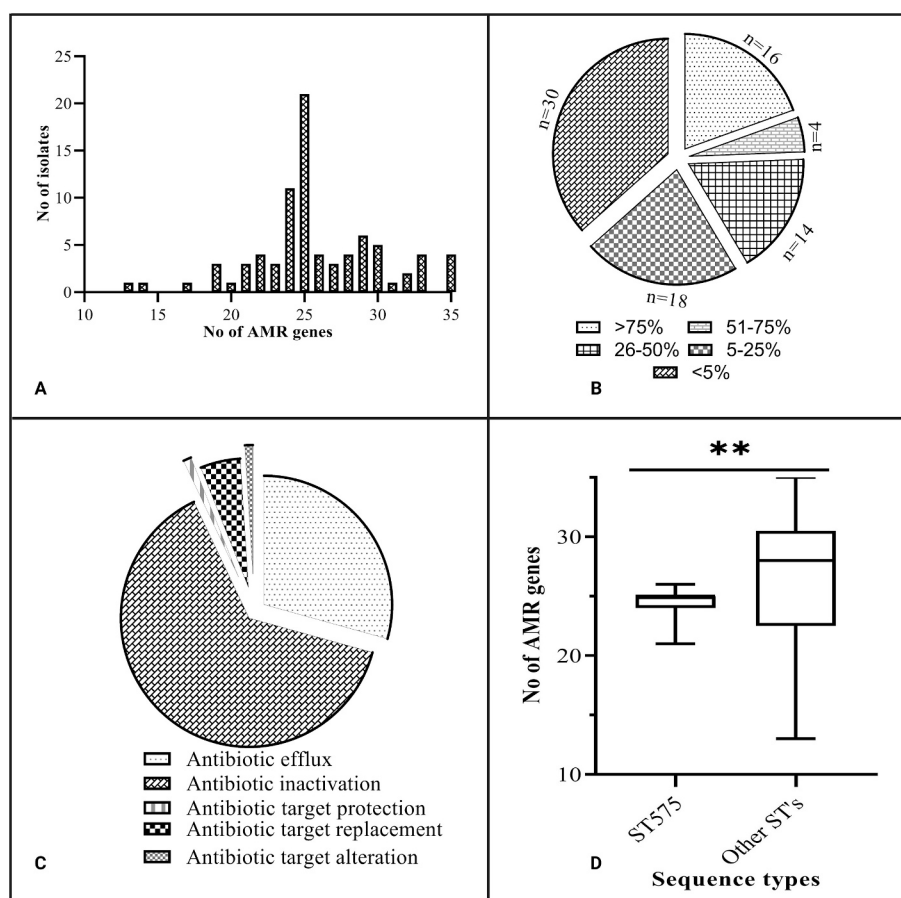


Fig. 4. Characterization of AMR genes in *A. baumannii* isolates. (A) per genome AMR contents, (2) categorization and distribution of MAR genes, (3) distribution of resistance mechanisms and (4) comparison of AMR genes among ST's.

replicates. The resulting phylogenetic tree was visualized using iTOL (Letunic and Bork, 2021).

2.10. Statistical analysis

All statistical analysis was performed using GraphPad Prism V8.0. A *P* value of <0.05 was considered statistically significant. The principal coordinate analysis (PCoA) of the strains based on the gene presence or absence matrix derived from pan-genome analysis was performed using PAST software (Hammer et al., 2005).

3. Results

3.1. Isolation and phenotypic characterization of *A. baumannii* S1C

A. baumannii S1C produced small, raised and pale colonies on MacConkey agar. It also showed positive catalase, negative oxidase, negative indole, negative methyl red, negative urease and non-hemolytic characteristics on different biochemical media. The isolate S1C exhibited resistance to 12 out of 13 (92.20%) used antibiotics except cotrimoxazole. Based on the phenotypic resistance pattern, S1C was found to be multidrug-resistant (MDR). Moreover, it also produced biofilm in microtiter plate assay. According to Stepanovic criteria, the extent of biofilm formation was moderate ($2 \times \text{ODc} < \text{ODs} \leq 4 \times \text{ODc}$; $\text{OD}_{\text{control}} = 0.08$ and $\text{OD}_{\text{isolate}} = 0.264$).

3.2. Genome analysis

The length of the assembled genome of S1C was 3875970 bp,

consisting of 251 contigs (>200 bp), with a GC content of 38.9% (Fig. 1). TYGS identified S1C as *A. baumannii* species which was further reconfirmed by Species Finder. Phylogenomic analysis based on genome-based distance phylogeny (GBDP) and codon tree-based method identified *A. baumannii* ATCC 19606 and *A. baumannii* ATCC 17978 as the closest relative of S1C, respectively (Supplementary Fig. 1). It also had 97.92% and 97.90% average nucleotide identity (ANI) with ATCC 19606 and ATCC 17978, respectively. According to the MLST analysis, S1C belonged to ST2 (Pasteur scheme) sequence type. Kaptive 2.0 assigned SC1 to K12 capsular type and OCL1 serotype.

Genome annotation of S1C revealed that it consisted of 3812 coding sequences involved in 301 subsystems. Most of the proteins were associated with metabolism of carbohydrates, nitrogenous compounds, amino acids and derivatives, nucleosides and nucleotides, co-factors, vitamins and pigments. Other notable subsystems include iron acquisition, virulence and defense, membrane transport, cell wall and capsules. We also identified a number of genes involved in the degradation of aromatic compounds including benzene, ethylbenzene, caprolactam, styrene, atrazine trinitrotoluene and others in the draft genome.

Scanning of the contigs in the draft genome against the CARD database detected the presence of 28 antimicrobial resistance genes. Antibiotic efflux was found to be the predominant resistance mechanism followed by target alteration, target protection and antibiotic inactivation. Mutations linked to fluoroquinolone resistance were detected in *gyrA* (S81L) and *parC* (S84L, V104I, D105E) genes. The identified acquired resistance genes comprised *blaOXA-23*, *blaOXA-66* (carbapenem), *armA* (amikacin, gentamicin), *blaADC* (cephalosporin), *mphE* (macrolide), *tet(B)* (tetracycline) etc. (Fig. 2). The assembled genome also contained 8 insertion sequences belonging to IS3, IS4, IS5, IS66,

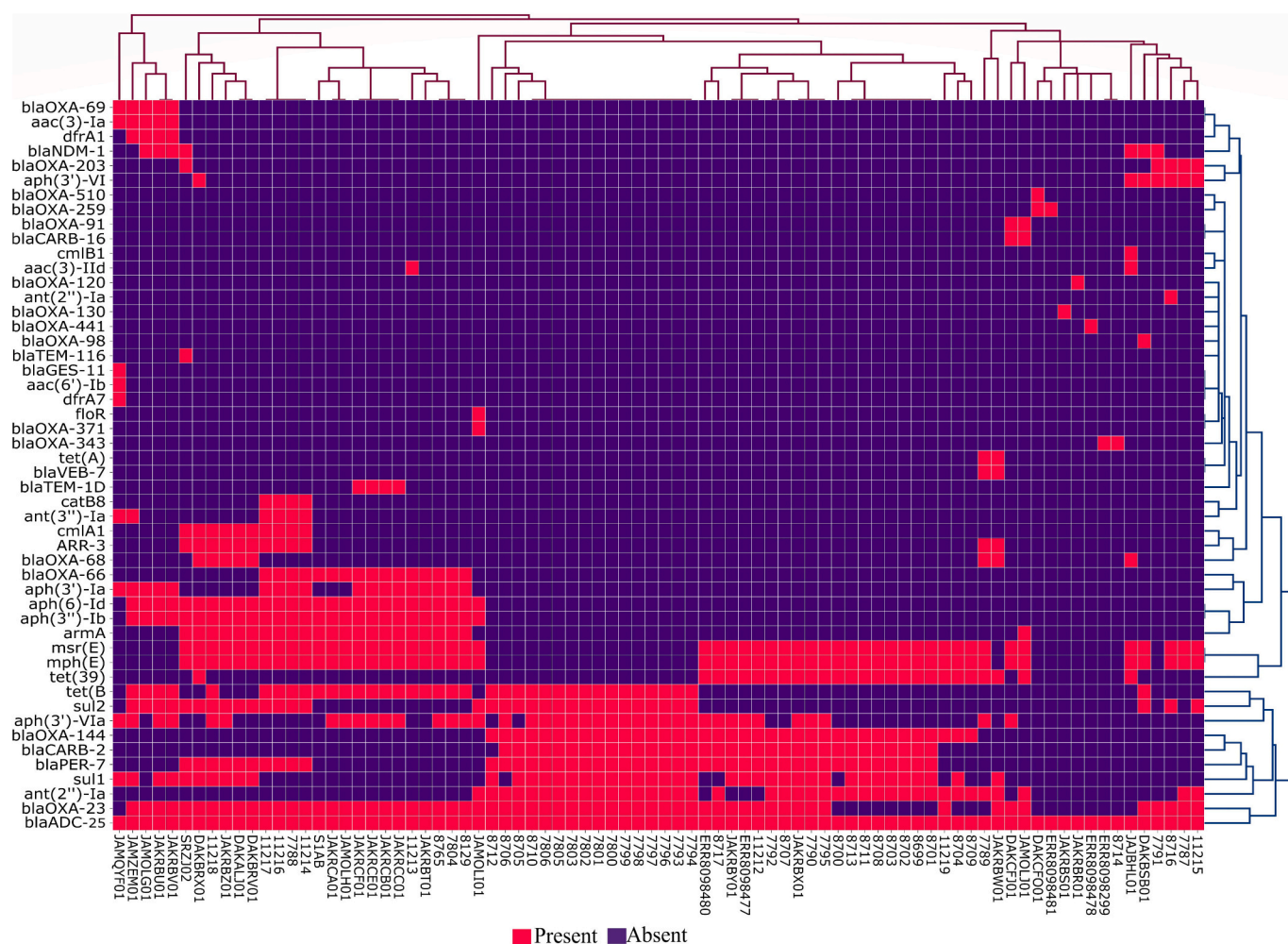


Fig. 5. Heatmap showing the presence or absence of acquired AMR genes within *A. baumannii* draft genomes.

IS91 and IS256 family. No plasmid sequences and integrons were identified in S1C genome. The search for virulence factors using VFDB database revealed the presence of numerous virulence factors involved in adherence to the host, biofilm formation, signal transduction, modulation of host immune response etc. Moreover, according to PathogenFinder, S1C was identified as a human pathogen with a probability score of 0.853 with 36.39% proteome coverage and 1302 pathogenic protein families. Some notable antimicrobial resistance and virulence genes of S1C with their product and association with resistance or virulence were listed in Table 1.

Assembly characteristics of retrieved genome

We retrieved 82 non-duplicated draft genomes from databases including S1C, that were reported from Bangladesh. None of them were found to be complete genome. The size of these genomes ranged between 3.6Mbp to 4.3Mbp with an average length of 3.9Mbp. The GC content varied from 38.75% to 39.32% with an average of 38.94%. The average of the number of contigs, L50, and N50 of the assemblies were calculated to be 138, 131392 bp and 18 respectively (Supplementary table 2).

3.3. Population analysis of *A. baumannii* in Bangladesh

Among 82 clinical strains of *A. baumannii*, 65.85% ($n = 54$) were isolated from patient's blood followed by tracheal aspirate (14.63%, $n = 12$) (Fig. 3). The rest of the samples included CSF ($n = 4$), wound swab ($n = 3$), urine ($n = 2$), throat swab ($n = 3$), endotracheal tube ($n = 2$) and unknown sample of clinical origin ($n = 2$). The MLST analysis using the

Pasteur scheme predicted a total of 17 sequence types (ST's). ST575 was found to be the predominant (45.12%, $n = 37$) clonal type followed by ST2 (19.51%, $n = 16$), ST1 and ST10 (6%, $n = 5$) (Fig. 4). The remaining ST's occurred at the number of ≤ 3 (3.6%). Kaptive based identification and typing of *A. baumannii* predicted that 82 draft genomes belong to 20 capsular (K locus) and 9 polysaccharide (O locus) types. The number of isolates belong to KL12 and KL13 capsular types were 21 and 20, respectively, whereas the outer (lipopolysaccharide) loci was predominated by OCL2 (56.09%, $n = 46$).

3.4. Profiling of antimicrobial resistance

In silico AMR profiling of retrieved genomes predicted the presence of a total of 82 intrinsic and acquired AMR genes. The average number of AMR genes per genome was 26. A maximum of 35 genes were detected in four genomes (Fig. 4A). We categorized 82 genes into five groups depending on their percentage of occurrence. >75% of strains carried 16 genes (*AmvA*, *abeM*, *abeS*, *adeG*, *adeH*, *adeI*, *adeJ*, *adeK*, *adeL*, *adeF*, *adeN*, *AbaF*, *AbaQ*, *adeS*, *adeC*, *blaOXA-23*) (Fig. 5B). These genes confer resistance through different mechanisms. In this study, antibiotic activation (63%, $n = 52$) was found to be the predominant resistance mechanism followed by antibiotic efflux (29%, $n = 24$), antibiotic target replacement (5%, $n = 4$), target protection, and target alteration (0.9%, $n = 1$) (Fig. 5C). A comparison of the size of the resistome between ST's suggested that the number of AMR genes in ST575 isolates was significantly ($p < 0.05$) higher than strains belong to other ST's.

Assembly based analysis of antibiotic resistome also predicted the

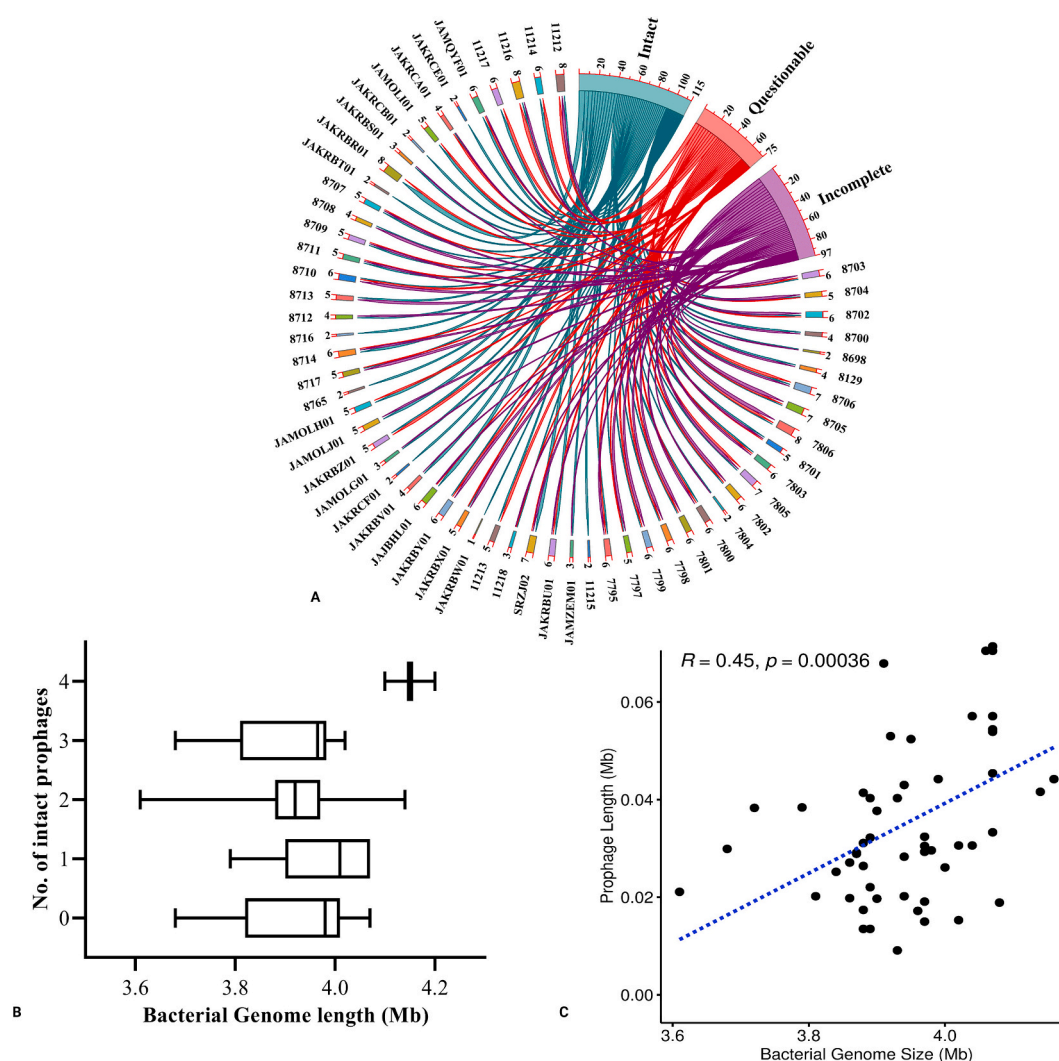


Fig. 6. Characteristics of prophages in *A. baumannii* genomes. (A) Distribution of prophages within genomes; (B) Relationship between bacterial genome length and the number of intact prophages; (C) Correlation between *A. baumannii* and prophages genome length.

presence of a wide variety of acquired AMR genes in these strains, conferring resistance to different classes of antibiotics. Among different beta-lactamase genes, all genomes (100%, $n = 82$) carried the *bla*ADC-25 gene followed by *bla*OXA-23 (75.6%, $N = 62$), *bla*PER-7 (53.65%, $N = 44$), *bla*OXA-144 (45.12%, $N = 37$) and *bla*CARB-2 (40.2%, $N = 33$) (Fig. 5). The proportion of isolates harbored *bla*OXA-66, *bla*NDM and *bla*TEM were 19.51% ($N = 16$), 8.53% ($N = 7$) and 6.09% ($N = 5$), respectively. The most abundant aminoglycoside resistance genes were *aph*(3')-VIa and *ant*(2'')-Ia₁₃ (48.78%, $N = 40$). Other aminoglycoside resistance genes included *aph*(3')-Ib and *aph*(6)-Id (32.92%, $N = 27$), *armA* (26.82%, $N = 22$), *aph*(3')-Ia (21.95%, $N = 18$), *aph*(3')-VI, (8.53%, $N = 7$), and others. 62.19% ($N = 52$) strains had both *mphE* and *msrE* genes that are associated with resistance to macrolide antibiotics. Among three different tetracycline resistance genes, *tet*(B) was found to be predominant (46.34%, $N = 38$) followed by *tet*(39) (31.7%, $N = 26$) and *tet*(A) (2.4%, $N = 2$). The frequency of *sul1*, *sul2* (sulfonamide resistance) and *catB8* (phenicol resistance) were 51.21% ($N = 42$), 37.8% ($N = 31$) and 4.87% ($N = 4$), respectively. We also identified fluoroquinolone resistance associated mutations in *gyrA* and *parC* genes. 91.46% ($N = 75$) and 86.58% ($N = 71$) of genomes had mutations in *gyrA* (S81L) and *parC* (S84L, V104I, D105E) gene, respectively. The percentage of strains that carried mutations in both gene was 85.26% ($N = 70$). In terms of ST wise distribution of AMR genes, we found that all ST2 isolates possessed *bla*OXA-23, *bla*OXA-66, *bla*ADC-25, *aph*(3')-Ib,

aph(6)-Id, *mphE*, *msrE* and *tet*(B) genes, whereas only *bla*ADC-25 was found in all isolates of ST575. Among 37 genomes of ST575, *bla*OXA-144, *ant*(2'')-Ia and *bla*CARB-2 occurred in 36, 33 and 32, respectively. Finally, based on *in silico* AMR analysis, all strains of *A. baumannii* could be designated as multidrug-resistant (MDR) as they had multiple genes presenting >3 different classes of antibiotics.

3.5. Detection of virulence genes

Putative virulence factors (VFs) genes predicted in 82 genomes were grouped under different categories such as adherence to host, biofilm formation, immune evasion, iron uptake and others. The number of VFs in genomes ranged from 24 to 34 with an average of 28. All the genome carried *barA*, *barB* (sideophore efflux system) *basACDFG*, *bauBDEF*, *entE* (acinetobactin biosynthesis and intake), *bfmR*, *bfmS* (signal transduction system), *plc* and *plcD* (phospholipase) genes (Supplementary fig. 2). Moreover, the gene *ompA* (outer membrane protein), *pgaA* (N-acetyl-D-glucosamine export porin), *csuA/B* (Csu pilus major pilin subunit) and *bap* (biofilm formation associated protein) occurred at a frequency of 87.80% ($N = 72$), 56.1% ($N = 46$), 45.12 ($N = 37$), and 26.82% ($N = 22$), respectively.

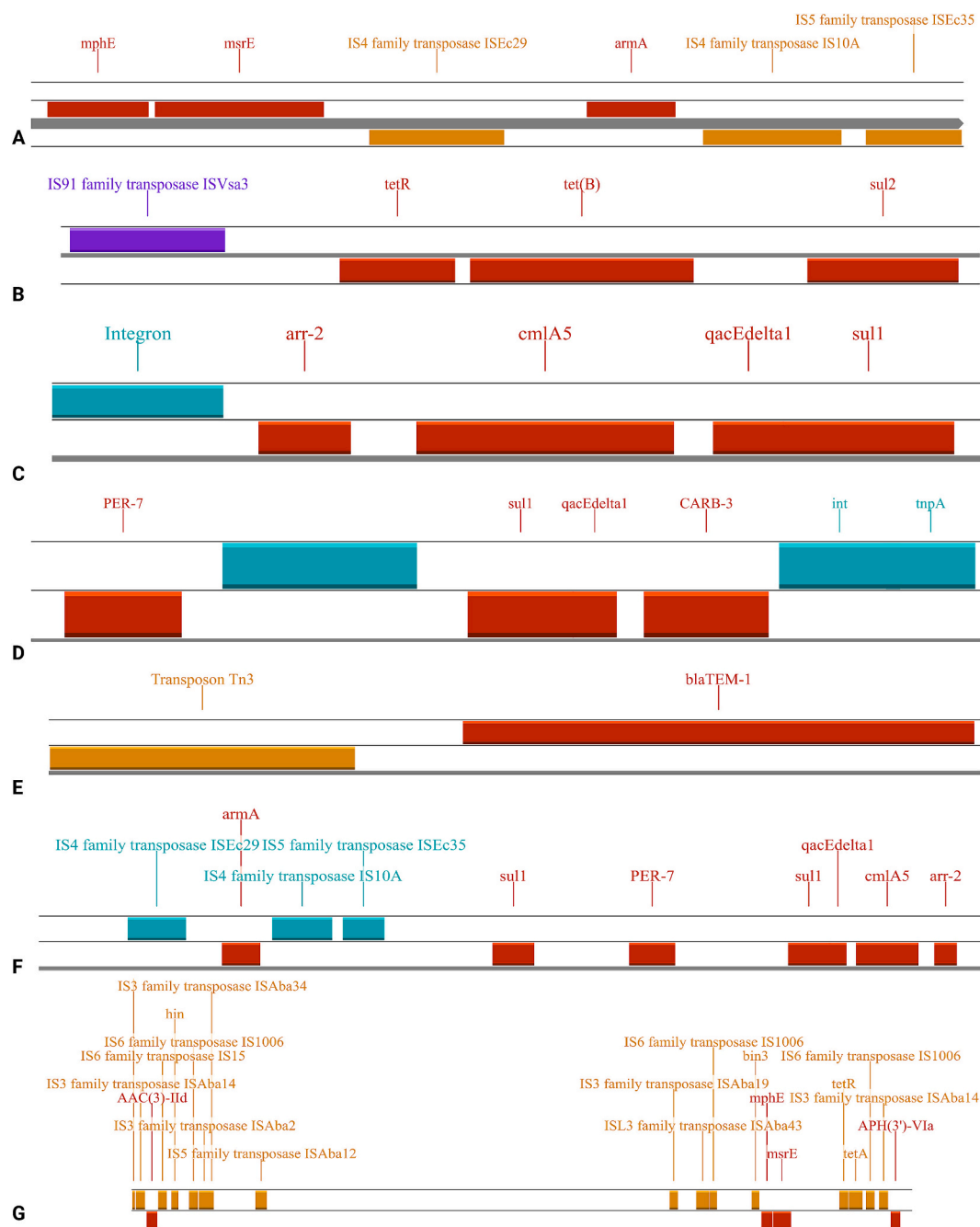


Fig. 7. Clusters of AMR genes associated with MGE. (A) *mphE*, *msrE* and *armA* associated with IS4, (B) *tet(B)* and *sul2* associated with ISVsa3, (C) *arr-2*, *cmlA5* and *sul1* associated with integron, (D) integron associated with *blaPER-7*, *sul1* and *blaCARB-3*, (E) *blaTEM* associated with Tn3, (F) multiple *amr* genes associated with IS4, and (G) *aac(S-IID)*, *mphE*, *msrE*, *tetA* and *aph(3)-Via* associated with IS3, IS5, IS6 family transposase.

3.6. Characterization of mobilome

We investigated the MGE's including plasmid sequences, insertion sequences, integrons and integrated prophage regions in 82 assembled genomes. According to PlasmidFinder, only one genome contained two plasmid sequences belonging to Col (MG828), ColRNAI types. Mobilome prediction by VRprofile2 showed that the size of the mobilome ranged from 3 to 33 with an average of 10 MGE's. The isolate *A. baumannii* strain MIN-005 (NZ_JAJBHL000000000.1) had a maximum of 33 MGE's consisting of 4 prophage regions, 2 conjugative transposons and 27 insertion sequences (ISE's). All genomes were analyzed using ISEscan for determining the number of ISE's and their type and distribution within genomes. The mean number of ISE's per genome was

approximately 8. IS5 was the mostly occurred IS family since 157 ISE's from this family found in 77 (93.90%) genomes. The number of ISE's belong to IS3, IS30, IS6, IS4, IS481, IS256 and IS91 families were 146, 77, 74, 50, 42, 30 and 22 respectively (Supplementary fig. 3). The frequency of ISE's per genome had no correlation with bacterial genome length. Furthermore, 9.7% ($N = 8$) genomes carried at least one or more integrons (In498, In83, In310, In153, In2-30, In304, and In474) at >90% identity and 60% coverage.

A total of 400 prophage regions were predicted in 82 assemblies, being distributed into 115 intact, 188 questionable and 97 incomplete prophages, suggesting a high prevalence of prophages in *A. baumannii* genomes (Fig. 6). We only considered intact phages for further analysis. There was a weak correlation ($R = 0.34$, $p = 0.0092$) between the

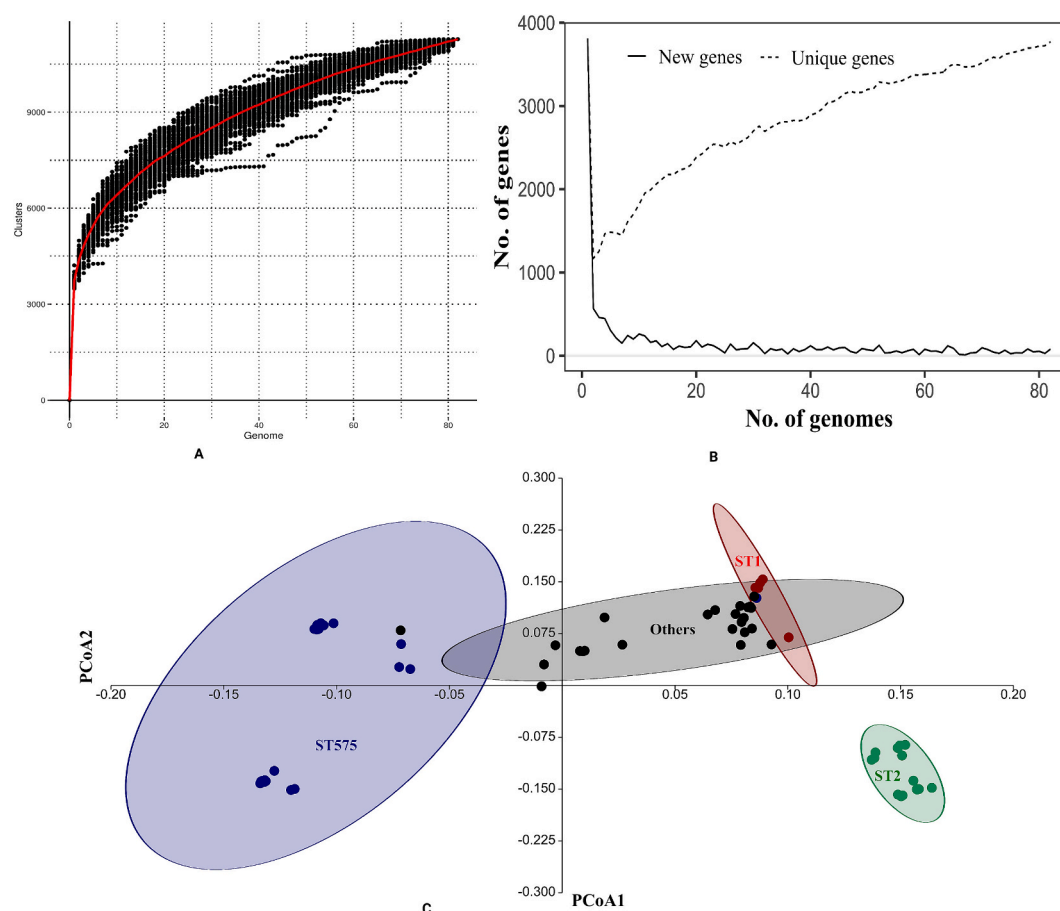


Fig. 8. Pan-genome analysis of 82 *A. baumannii* genomes from Bangladesh. (A) Pan-genome rarefaction curve, (B) the new vs unique genes curve, and (C) Principal Coordinate Analysis (PCoA) based on the presence/absence binary matrix of the pan genome.

number of prophages per genome and bacterial genome length. These intact phages including PHAGE_Acinet_YMC11/11/R3177, Acinet_Bphi_B1251, Acinet_Bphi_B1251, Salmon_SSU5, Acinet_vB_AbaS_TRS1 and others, were distributed in 60 of 82 (73.17%) *A. baumannii* isolates with 1 to 4 prophages per isolates. The genome length of intact prophages ranged from 9.1 kb to 71.4 kb with an average length of 34 kb. We also observed a moderate positive correlation between the genome length of host and prophages. Scanning of prophage sequences using ABRicate against CARD and ResFinder database did not yield any hit for AMR genes.

We predicted the association between AMR genes and MGEs along with their origin (chromosome or plasmid) using VRprofile2 (Fig. 7). The co-occurrence of IS4 and IS5 family transposase with macrolide resistance associated *mphE*, *msrE* and *armA* gene in the same contig was observed in 18 isolates (21.95%). 15 (18.29%) genomes had ISVsa3 element containing *sul2* and *tet(B)* gene. Our analysis also detected a 7.2 kb region of plasmid sequence co-harboring IS26 with *blaCARB-2*, *qacE*, *sul1*, and *blaPER-7* genes 8 (9.75%) isolates. Only 3 isolates carried Tn3 transposase associated with *blaTEM-1D* gene in plasmid sequences. Plasmid replicons that carried multiple IS elements and AMR genes were also detected across genomes. For example, one isolate (pubMLST ID: 11218) possessed a 16.7kb region containing IS cluster (ISEc29, IS10A, and ISEc35) with 6 AMR genes (*armA*, *sul1*, *blaPER-7*, *qacE*, *cmlA1*, *ARR-3*) that showed 100% identity with several *A. baumannii* plasmids Pb11911, pIOMTU433, p1KSK6, and p1KSK1. Similarly, the strain MIN-005 carried two IS clusters (3 IS3, 2 IS6 and 1 IS1 family transposases in each cluster) with several AMR genes (*aac(3)-IId*, *cmlB1*, *mphE*, *msrE*, *tet(A)*, *aph(3')-VI*) in the same contig.

3.7. Pan-genome analysis

The pan-genome consisted of 12250 genes, being distributed into 1798 (14.67%) core (99% ≤ strains ≤ 100%), 669 soft cores (5.46%) (95% ≤ strains < 99%), 2310 shell (18.85%) (15% ≤ strains < 95%) and 7473 (61%) cloud genes (0% ≤ strains < 15%). The Pan-genome rarefaction curve experienced a logarithmic increase in the number of pan-genome clusters with the addition of new genomes against a conserved core gene (Fig. 8). Similarly, as more genomes were added, the number of new genes remained almost constant whereas the number of unique genes increased sharply. Overall, the results indicated the “open” structure of *A. baumannii* pangenome. PCoA-based clustering-based presence/absence of gene supported the clonal nature of the isolates as evident by the clustering patterns of the isolates belonging to the same sequence types. In addition, the average nucleotide identity (ANI) value among the genomes ranged between 97% to 99%, indicating a close genomic relationship among the *A. baumannii* isolates (Supplementary fig. 4).

3.8. SNP-based phylogenetic characterization

The SNP-based phylogenetic tree grouped 82 isolates into three distinct clades. Clade 1 can be divided into two main subclades that consisted of 6 genomes. 31 genomes formed Clade 2 that also had two main subclades. The genome of our isolated bacteria S1C occurred in this clade. *A. baumannii* strains from ST575, ST10 and ST23 sequence types clustered together in clade 3 in which each sequence types formed separate subclades (Fig. 9). Interestingly, ST1, ST2 and ST575 were found to be predominant in Clade 1, Clade 2 and Clade 3, respectively

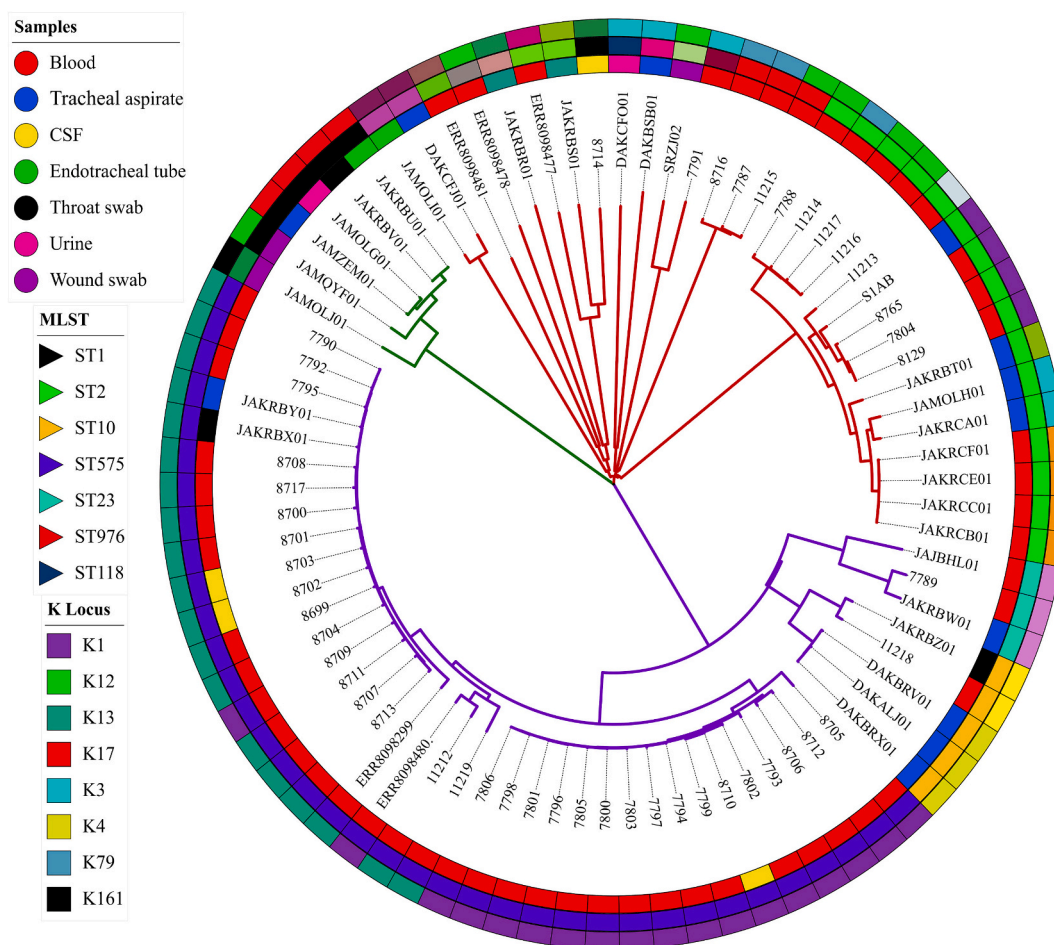


Fig. 9. Core genome SNP based phylogenetic tree of 82 *A. baumannii* genomes from Bangladesh (Here, Clade 1, Clade 2, and Clade 3 are colored green, red and blue, respectively). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and also formed major monophyletic clades. In addition, the constructed phylogenetic showed that there was no isolation source specific clustering of *A. baumannii*.

4. Discussion

Previous studies on AMR in *A. baumannii* were primarily based on phenotypic and genotypic characterization, as well as infection frequency measurements. While these studies provided important information on the prevalence of *A. baumannii* in hospitalized patients and trends in resistance and virulence, the molecular epidemiology and mechanisms of resistance remained unclear. In recent years, several studies in Bangladesh have employed WGS for isolate identification, sequence typing, and elucidation of AMR patterns in *A. baumannii*. However, these individual studies could not provide a comprehensive depiction of the AMR landscape in *A. baumannii*. To address this gap, we conducted genome sequencing of *A. baumannii* S1C and compiled reported *A. baumannii* genomes from Bangladesh to comprehensively characterize population distribution, AMR, virulence factors, gene transmissibility, pan-genome, and phylogeny within the population.

Continued surveillance and monitoring efforts are urgently needed to track the evolution of AMR. Our isolated strain S1C was multidrug-resistant, a moderate biofilm-former that belong to sequence type ST2. Isolates representing the ST2 clonal type usually exhibit MDR characteristics, strong biofilm formation activity, and high pathogenicity, thus considered one of the high-risk clones of *A. baumannii* causing outbreaks worldwide (Morgado et al., 2023). Therefore, infections caused by ST2 *A. baumannii* are difficult to control, leading to high mortality (Yu et al.,

2021). Our findings on the characteristics of S1C were consistent with other studies that also reported similar observations on ST2 isolates in clinical and environmental settings in Bangladesh (Farzana et al., 2022; Rahman et al., 2022). In this study, using ABRicate with the Plasmid-Finder database did not yield any results for the presence of plasmid sequences in the draft genome. This highlights a limitation of this approach, as it relies on known plasmid sequences within the database, potentially missing novel or highly divergent plasmids (Carattoli et al., 2014). Further exploration of S1C genome also identified several key genetic determinants associated with AMR, specifically, mutations in the *gyrA* and *parC* genes (fluoroquinolone resistance), as well as the presence of the *OXA-23* and *armA* genes (carbapenem and aminoglycoside resistance). These findings are supported by a study which reported an association of *A. baumannii* ST2 isolates with *blaOXA-23* and *armA* genes in Bangladesh (Farzana et al., 2022). Furthermore, a study from China demonstrated the hospital dissemination of MDR *A. baumannii* strains carrying *blaOXA-23*, *armA*, and mutations in quinolone resistance-determining regions (QRDR) (Shen et al., 2016). S1C's similar resistance profile and biofilm-forming capability suggested its persistence and transmission in hospital environments. We also observed high genotype-phenotype concordance, with the majority of identified AMR genes correlating with observed phenotypic resistance patterns.

The association between ST's and high rates of AMR including MDR and XDR, is well-established (Hussain et al., 2023). MLST analysis revealed a total of 17 ST's among the strains, highlighting the genetic heterogeneity present within the Bangladeshi *A. baumannii* population. ST575 was identified as the most prevalent clonal type, suggesting a

potential dominance of ST575 strains within the Bangladeshi setting, which may have important epidemiological implications in terms of transmission dynamics and infection control strategies. Previous studies have also reported the occurrence of this ST in other Asian countries, including India, Vietnam, and Myanmar (Kumkar et al., 2022; Tada et al., 2020, 2015). Next to ST575, ST2 types constituted a substantial proportion, with 16 strains identified within the dataset. The international clone ST2 remains the most prevalent sequence type identified across all sequenced *A. baumannii* genomes worldwide to date (Levy-Blitchtein et al., 2018). The predominance of ST2 has been reported in many parts of the world, including China, Taiwan, Thailand, Germany, Brazil, Peru, and Sudan (Al-Hassan et al., 2021; Chuang et al., 2019; Eigenbrod et al., 2019; Levy-Blitchtein et al., 2018; Yang et al., 2018). We also observed the occurrence of other STs, notably ST1, ST10, ST23, and ST976, at relatively lower frequencies. Overall, the analysis of clonal types predicted genomic diversity in the Bangladeshi population of clinical *A. baumannii*, with heterogeneous distribution. Such diversity highlights the significance of genomic surveillance in monitoring the epidemiology and evolution of *A. baumannii* in Bangladesh.

The ability of *A. baumannii* to develop or acquire resistance through various mechanisms renders it among the most formidable nosocomial pathogens. Our *in silico* analysis identified 82 different AMR genes in Bangladeshi *A. baumannii* strains, with antibiotic inactivation being the predominant resistance mechanism. All isolates of the ST2 clonal type carried the *blaOXA-23* gene, with an overall frequency of 75.6%. These findings are comparable to a study from India that reported 72.34% of *A. baumannii* isolates harbored this gene, including 14 out of 18 (77%) ST2 isolates [7]. Similarly, a recent study from Thailand mentioned that all ST2 isolates of carbapenem-resistant *A. baumannii* (CRAB) had *blaOXA-23* (Chukamnerd et al., 2022). Such a high occurrence of *blaOXA-23* in ST2 isolates highlights the contribution of a specific sequence type to the dissemination of carbapenem resistance. In this study, all except one isolate from ST575 had the *blaOXA-144* gene. Tada et al. also found similar results in ST575 isolates of *A. baumannii* isolated from the ICU of a hospital in Vietnam (Tada et al., 2015). Our analysis showed that ST575 carried a significantly higher number of AMR genes compared to other STs. Other notable beta-lactamases included *blaPER-7*, *blaCARB-2*, and *blaOXA-66*. *blaPER-7* is an extended-spectrum beta-lactamase (ESBL) with higher hydrolytic activity against cephalosporins and aztreonam, whereas *blaOXA-66* is the most prevalent carbapenem-hydrolyzing beta-lactamase in *A. baumannii* worldwide (Bonnin et al., 2011; Hu et al., 2007). In this study, all 16 genomes of ST2 exhibited the co-existence of *blaOXA-23* and *blaOXA-66*. Many studies have shown an association between the co-occurrence of these genes and elevated resistance rates in clinical *A. baumannii*, providing a scientific basis for the failure of beta-lactam therapy (Al-Hassan et al., 2021; Li et al., 2023). Apart from beta-lactamases, *gyrA* subunit (S81L) and *parC* subunit (S84L, V104I, and D105E) mutations were detected in 85.36% of strains, including all from ST2. Likewise, these mutations were found to be prevalent in MDRAB, specifically in ST2 populations in other countries (Baleivuanalala et al., 2023; Morgado et al., 2023). Our analysis identified *mphE* and *msrE* as the most abundant macrolide resistance genes, which confer resistance to macrolides by inactivation and modification of the target site, respectively (Kyriakidis et al., 2021). Moreover, while comparing our findings with other studies, we noted different degrees of variation in the abundance of tetracycline (*tet(B)*), chloramphenicol (*catB8*), and aminoglycoside resistance genes (*aph(3')-Via*, *ant(2'')-Ia*, *aph(3'')-Ib*, *aph(6)-Id*, *armA*, and *aph(3')-Ia*) (Cui et al., 2023). These differences might be attributed to the number of samples studied, geographical variations, evolutionary dynamics, selective pressure in the environment, and other factors. According to the presence of genes conferring resistance to three or more classes of antibiotics, all the analyzed strains were classified as MDR. Overall, the size and composition of the resistome indicated versatility and diversity in AMR genes in *A. baumannii*, with the ability to withstand a wide variety of antibiotics in Bangladesh.

A. baumannii is equipped with a specialized arsenal of virulence factors that offer advantages for its fitness across different stages of pathogenesis (Shadan et al., 2023). We did not observe any notable variations in the virulence factor (VF) contents among *A. baumannii* isolates in Bangladesh. The potential VFs identified are mainly associated with modulation of host factors, adherence, effector delivery, nutrient acquisition, host immune response, and others. All genomes had VF genes for iron acquisition, acinetobactin biosynthesis, immune evasion, and phospholipase synthesis, indicating the isolates' pathogenicity and infection severity (Liu et al., 2018). Varying degrees of occurrence of *ompA*, *bap*, and *csu* pili genes were found, which are known to play roles in motility, adhesion, and biofilm formation. The biofilm-associated protein Bap mediates biofilm formation and maturation, and its role in intercellular adhesion is thought to be associated with nosocomially acquired device-related *A. baumannii* infections (Loehfelm et al., 2008). *Csu* pilus is a crucial virulence factor for biofilm formation and also promotes disease progression by facilitating bacterial adhesion to epithelial cells (Ahmad et al., 2023). Another important VF of *A. baumannii* is *OmpA*, which has been associated with antibiotic resistance, adherence and invasion of host cells, and stimulation of the immune response (Uppalapati et al., 2020). Our analysis found that 87.80% of the genomes harbored the *ompA* gene, indicating its widespread presence among *A. baumannii* strains in Bangladesh. This finding is consistent with a recent study that reported an *ompA* prevalence of 86% among *A. baumannii* isolates from patients with burns and ventilator-associated pneumonia in Iran (Khoshnood et al., 2023). Overall, Bangladeshi *A. baumannii* strains possessed a rich repertoire of virulence genes required for their survival strategies and pathogenesis.

Most reported cases of transmissible antibiotic resistance in *A. baumannii* have been linked to mobile genetic elements (MGEs) such as insertion sequences (ISs), transposons, and plasmids, underlying the significance of MGEs in AMR transmission (Chakravarty, 2020). In our analysis of the *A. baumannii* genomes, we identified insertion sequences from various families, with IS5 being the most prevalent, which is consistent with previous studies (Chamoun et al., 2021). The investigation on the detection of prophage sequences indicated a high prevalence of prophages in the *A. baumannii* genome dataset. Prophages are another type of MGE that have been linked to bacterial diversity and evolution as vehicles of horizontal gene transfer (HGT), and they may have significant effects on the fitness and pathogenicity of bacteria (Costa et al., 2018). We observed that 74.07% of genomes harbored intact prophages; however, these prophages did not contain any genes associated with AMR or virulence. This finding contrasts with the results reported by Costa et al., who observed a similar occurrence of prophages (74.1%) containing numerous virulence genes (Costa et al., 2018). The completeness of the bacterial genome, the number of studied genomes, the length of intact phages, and the rate of HGT might be responsible for this discrepancy. Although this study lacks data on the abundance and type of plasmids in *A. baumannii*, the search for MGE-associated AMR genes revealed their co-occurrence in contigs of plasmid origin. In a notable proportion of strains, we observed the presence of the *mphE*, *msrE*, and *armA* genes within plasmid sequences, with *armA* flanked by ISEC29 and ISEC35. Camargo et al. reported a similar scenario with *armA* and *OXA-23* co-producing ST2/IC2 *A. baumannii* in Brazil during the COVID-19 pandemic (Camargo et al., 2022). In alignment with other studies, we also observed an association of the ISVsa3 element with *sul2* and *tet(B)* genes (Hu et al., 2024; Mehrotra et al., 2023). Plasmids carrying AMR genes have the ability to confer resistance to multiple classes of antibiotics, limiting treatment options and increasing the risk of treatment failure. Here, IS26-bound *blaCARB-2*, *qacE*, *sul1*, and *blaPER-7* were detected in a 7.2 kb plasmid sequence of 8 isolates. Another plasmid sequence had the co-carriage of an IS cluster with several AMR genes, conferring resistance to aminoglycoside, sulfonamide, beta-lactam, and phenicol classes of antibiotics. *A. baumannii* plasmids co-harboring different AMR genes were evident in earlier studies (Jeon et al., 2023; Karah et al., 2020; Sánchez-Urtaza et al., 2023). These

findings provide an overview of the mobilome structure of *A. baumannii* in Bangladesh followed by elucidating their association with AMR and transmission characteristics.

Pan-genome analysis elucidates genomic diversity within bacterial species (Tettelin et al., 2008). The pan-genome of 82 Bangladeshi *A. baumannii* strains showed 61% accessory genes, suggesting genomic flexibility for acquiring and losing genetic elements over time. A Thai study found 68.68% accessory/cloud genes in 221 isolates. High genomic plasticity was evident from rapidly increasing pan-genes relative to constant core genes. The rise in unique genes with additional genomes reflects distinct genetic diversity within Bangladeshi *A. baumannii*. Previous studies noted genomic variability among persistent *A. baumannii* strains in Italy, Argentina, Australia, and Thailand (Chukamnerd et al., 2022; Gaiarsa et al., 2019; Mussi et al., 2011; Valenzuela et al., 2007). The open pan-genome indicates ability for gene acquisition/loss, facilitating niche adaptation and potentially increasing pathogenicity (Imperi et al., 2011). Phylogenetic analysis revealed three distinct clades within the Bangladeshi *A. baumannii* population. Interestingly, isolates of the same sequence type (ST) clustered exclusively within individual clades, suggesting phylogenetic clustering by genetic similarity, as observed in Malaysian and Thai populations (Chukamnerd et al., 2022; Muzahid et al., 2023). Clade 3 spanned a broader ST range (ST575, ST23, ST10), indicating more diverse genetic composition or lineage convergence. Principal coordinates analysis further affirmed sequence type-wise clustering of Bangladeshi *A. baumannii* populations.

5. Conclusion

This study provides a comprehensive scenario of the population structure, resistance genes, virulence factors, pan-genome, and phylogeny of clinical *A. baumannii* in Bangladesh. One limitation is the lack of consideration of the genomes' collection year, which might provide insights into shifts in population distribution and changes in AMR trends over time. Despite this limitation, the study yielded crucial insights into the genetic determinants of *A. baumannii*, contributing to our understanding of this pathogen's epidemiology and pathogenicity in clinical settings. This understanding is necessary for developing targeted intervention strategies to combat this emerging threat. Based on our findings, notably the circulation of high-risk clones, diversity in AMR genes, richness of virulence factors, MGE-associated transmission of resistance genes, and the genomic plasticity of *A. baumannii* in Bangladesh, we recommend using genome sequencing for surveillance of this notorious pathogen and subsequently controlling its emergence and transmission in Bangladesh.

CRedit authorship contribution statement

Md Abu Sayem Khan: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Sudipta Chowdhury Chaity:** Writing – original draft, Methodology, Formal analysis, Data curation. **Md Arman Hosen:** Writing – original draft, Methodology, Formal analysis. **Sabita Rezwana Rahman:** Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare no conflict of interest.

Data availability

The draft genome sequence of *A. baumannii* S1C was deposited into NCBI GenBank under the accession number JAZDCM000000000.1.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.meegid.2024.105656>.

References

- Ahmad, I., Nadeem, A., Mushtaq, F., Zlatkov, N., Shahzad, M., Zavialov, A.V., Wai, S.N., Uhlir, B.E., 2023. *Csu* pili dependent biofilm formation and virulence of *Acinetobacter baumannii*. *npj Biofilms Microbiomes* 9, 101.
- Alcock, B.P., Raphenya, A.R., Lau, T.T.Y., Tsang, K.K., Bouchard, M., Edalatmand, A., Huynh, W., Nguyen, A.-L.V., Cheng, A.A., Liu, S., 2020. CARD 2020: antibiotic resistance surveillance with the comprehensive antibiotic resistance database. *Nucleic Acids Res.* 48, D517–D525.
- Al-Hassan, L., Elbadawi, H., Osman, E., Ali, S., Elhag, K., Cantillon, D., Wille, J., Seifert, H., Higgins, P.G., 2021. Molecular epidemiology of carbapenem-resistant *Acinetobacter baumannii* from Khartoum state, Sudan. *Front. Microbiol.* 12, 628736.
- Arndt, D., Grant, J.R., Marcu, A., Sajed, T., Pon, A., Liang, Y., Wishart, D.S., 2016. PHASTER: a better, faster version of the PHAST phage search tool. *Nucleic Acids Res.* 44, W16–W21.
- Aziz, R.K., Bartels, D., Best, A.A., DeJongh, M., Disz, T., Edwards, R.A., Formsma, K., Gerdes, S., Glass, E.M., Kubal, M., 2008. The RAST server: rapid annotations using subsystems technology. *BMC Genomics* 9, 1–15.
- Baleivanualala, S.C., Isaia, L., Devi, S.V., Howden, B., Gorrie, C.L., Matanitobua, S., Sharma, S., Wilson, D., Kumar, S., Maharaj, K., 2023. Molecular and clinical epidemiology of carbapenem resistant *Acinetobacter baumannii* ST2 in Oceania: a multicountry cohort study. *Lancet Reg. Heal. Pacific* 40.
- Bankevich, A., Nurk, S., Antipov, D., Gurevich, A.A., Dvorkin, M., Kulikov, A.S., Lesin, V. M., Nikolenko, S.I., Pham, S., Pribelski, A.D., 2012. SPAdes: a new genome assembly algorithm and its applications to single-cell sequencing. *J. Comput. Biol.* 19, 455–477.
- Bolger, A.M., Lohse, M., Usadel, B., 2014. Trimmomatic: a flexible trimmer for Illumina sequence data. *Bioinformatics* 30, 2114–2120.
- Bonnin, R.A., Potron, A., Poirel, L., Lecuyer, H., Neri, R., Nordmann, P., 2011. PER-7, an extended-spectrum β -lactamase with increased activity toward broad-spectrum cephalosporins in *Acinetobacter baumannii*. *Antimicrob. Agents Chemother.* 55, 2424–2427.
- Bortolaia, V., Kaas, R.S., Ruppe, E., Roberts, M.C., Schwarz, S., Cattoir, V., Philippon, A., Allesoe, R.L., Rebelo, A.R., Florensa, A.F., 2020. ResFinder 4.0 for predictions of phenotypes from genotypes. *J. Antimicrob. Chemother.* 75, 3491–3500.
- Brown, E.D., Wright, G.D., 2016. Antibacterial drug discovery in the resistance era. *Nature* 529, 336–343.
- Camargo, C.H., Yamada, A.Y., Nagamori, F.O., de Souza, A.R., Tiba-Casas, M.R., de Moraes França, F.A., Porto, M.H.T.N., de Lima Garzon, M.L., Higgins, P., Madalosso, G., 2022. Clonal spread of ArmA and OXA-23-coproducing *Acinetobacter baumannii* international clone 2 in Brazil during the first wave of the COVID-19 pandemic. *J. Med. Microbiol.* 71, 1509.
- Carattoli, A., Hasman, H., 2020. PlasmidFinder and in silico pMLST: identification and typing of plasmid replicons in whole-genome sequencing (WGS). *Horiz. Gene Transf. Methods Protoc.* 285–294.
- Carattoli, A., Zankari, E., García-Fernández, A., Voldby Larsen, M., Lund, O., Villa, L., Møller Aarestrup, F., Hasman, H., 2014. In silico detection and typing of plasmids using PlasmidFinder and plasmid multilocus sequence typing. *Antimicrob. Agents Chemother.* 58, 3895–3903.
- Chakravarty, B., 2020. Genetic mechanisms of antibiotic resistance and virulence in *Acinetobacter baumannii*: background, challenges and future prospects. *Mol. Biol. Rep.* 47, 4037–4046.
- Chamoun, S., Welander, J., Martis-Thiele, M.-M., Ntzouni, M., Claesson, C., Vikström, E., Turkina, M.V., 2021. Colistin dependence in extensively drug-resistant *Acinetobacter baumannii* strain is associated with IS Ajo2 and IS Aba13 insertions and multiple cellular responses. *Int. J. Mol. Sci.* 22, 576.
- Chen, L., Yang, J., Yu, J., Yao, Z., Sun, L., Shen, Y., Jin, Q., 2005. VFDB: a reference database for bacterial virulence factors. *Nucleic Acids Res.* 33, D325–D328.
- Chuang, Y.-C., Cheng, A., Sun, H.-Y., Wang, J.-T., Chen, Y.-C., Sheng, W.-H., Chang, S.-C., 2019. Microbiological and clinical characteristics of *Acinetobacter baumannii* bacteremia: implications of sequence type for prognosis. *J. Infect.* 78, 106–112.
- Chukamnerd, A., Singkhamanan, K., Chongsuvivatwong, V., Palittapongarnpim, P., Doi, Y., Pomwised, R., Sakunrang, C., Jeenkeawpim, K., Yingkajorn, M., Chusri, S., 2022. Whole-genome analysis of carbapenem-resistant *Acinetobacter baumannii* from clinical isolates in southern Thailand. *Comput. Struct. Biotechnol. J.* 20, 545–558.
- Cosentino, S., Voldby Larsen, M., Møller Aarestrup, F., Lund, O., 2013. PathogenFinder-distinguishing friend from foe using bacterial whole genome sequence data. *PLoS One* 8, e77302.
- Costa, A.R., Monteiro, R., Azeredo, J., 2018. Genomic analysis of *Acinetobacter baumannii* prophages reveals remarkable diversity and suggests profound impact on bacterial virulence and fitness. *Sci. Rep.* 8, 15346.
- Cui, F., Liu, L., Miao, X., Qian, H., Lu, S., Tian, J., Qiao, G., Shao, B., Li, Q., Zhang, R., 2023. Epidemiological and comparative genomic analyses of multidrug-resistant *Acinetobacter baumannii* collected between 2020 and 2022 in Liaocheng City, Shandong Province, China. *J. Biosaf. Biosecurity* 5, 60–66.
- Eigenbrodt, T., Reuter, S., Gross, A., Kocer, K., Günther, F., Zimmermann, S., Heeg, K., Mutters, N.T., Nurjadi, D., 2019. Molecular characterization of carbapenem-resistant *Acinetobacter baumannii* using WGS revealed missed transmission events in Germany from 2012–15. *J. Antimicrob. Chemother.* 74, 3473–3480.

- Farzana, R., Swedberg, G., Giske, C.G., Hasan, B., 2022. Molecular and genetic characterization of emerging carbapenemase-producing *Acinetobacter baumannii* strains from patients and hospital environments in Bangladesh. *Infect. Prev. Pract.* 4, 100215.
- Gaiarsa, S., Bitar, I., Comandatore, F., Corbella, M., Piazza, A., Scaltriti, E., Villa, L., Postiglione, U., Marone, P., Nucleo, E., 2019. Can insertion sequences proliferation influence genomic plasticity? Comparative analysis of *Acinetobacter baumannii* sequence type 78, a persistent clone in Italian hospitals. *Front. Microbiol.* 10, 2080.
- Gao, Y., Li, H., Zhao, C., Li, S., Yin, G., Wang, H., 2024. Machine learning and feature extraction for rapid antimicrobial resistance prediction of *Acinetobacter baumannii* from whole-genome sequencing data. *Front. Microbiol.* 14, 1320312.
- Gurevich, A., Savelyev, V., Vyahhi, N., Tesler, G., 2013. QUAST: quality assessment tool for genome assemblies. *Bioinformatics* 29, 1072–1075.
- Hammer, O., Harper, D.A.T., Ryan, P.D., 2005. PAST—Palaeontological statistics, ver. 1.34. Paleontol. Museum Oslo.
- Hu, M., Sun, H., Xu, Y., Xu, X., 2024. Antimicrobial susceptibility and genetic characteristics of multi-drug resistant *Acinetobacter baumannii* isolates in Northwest China. *Front. Microbiol.* 15, 1293725.
- Hu, W.S., Yao, S.-M., Fung, C.-P., Hsieh, Y.-P., Liu, C.-P., Lin, J.-F., 2007. An OXA-66/OXA-51-like carbapenemase and possibly an efflux pump are associated with resistance to imipenem in *Acinetobacter baumannii*. *Antimicrob. Agents Chemother.* 51, 3844–3852.
- Hua, X., Liang, Q., Deng, M., He, J., Hong, W., Leptihn, S., Yu, Y., Chen, H., 2021. BacAnt: a combination annotation server for bacterial DNA sequences to identify antibiotic resistance genes, integrons, and transposable elements. *Front. Microbiol.* 12, 649969.
- Hussain, A., Mazumder, R., Ahmed, A., Saima, U., Phelan, J.E., Campino, S., Ahmed, D., Asadulghani, M., Clark, T.G., Mondal, D., 2023. Genome dynamics of high-risk resistant and hypervirulent *Klebsiella pneumoniae* clones in Dhaka, Bangladesh. *Front. Microbiol.* 14, 1184196.
- Imperi, F., Antunes, L.C.S., Blom, J., Villa, L., Iacono, M., Visca, P., Carattoli, A., 2011. The genomics of *Acinetobacter baumannii*: insights into genome plasticity, antimicrobial resistance and pathogenicity. *IUBMB Life* 63, 1068–1074.
- Islam, N., Ahmed, D., Ahsan, N., Ahsan, K., Yasmin, M., 2023. Phenotypic-genotypic features of MDR *Pseudomonas aeruginosa* and *Acinetobacter baumannii* from Dhaka, Bangladesh. *Bioresearch Commun.* 9, 1276–1284.
- Jacobs, A.C., Blanchard, C.E., Catherman, S.C., Dunman, P.M., Murata, Y., 2014. An ribonuclease T2 family protein modulates *Acinetobacter baumannii* abiotic surface colonization. *PLoS One* 9, e85729.
- Jahan, M.I., Rahaman, M.M., Hossain, M.A., Sultana, M., 2021. Draft genome sequence of a carbapenem-resistant clinical *Acinetobacter baumannii* revealing co-existence of four classes of β -lactamases. *J. Glob. Antimicrob. Resist.* 27, 329–331.
- Jeon, J.H., Jang, K.-M., Lee, J.H., Kang, L.-W., Lee, S.H., 2023. Transmission of antibiotic resistance genes through mobile genetic elements in *Acinetobacter baumannii* and gene-transfer prevention. *Sci. Total Environ.* 857, 159497.
- Jolley, K.A., Bray, J.E., Maiden, M.C.J., 2018. Open-access bacterial population genomics: BIGSdb software, the PubMLST.org website and their applications. *Wellcome Open Res.* 3.
- Karah, N., Khalid, F., Wai, S.N., Uhlin, B.E., Ahmad, I., 2020. Molecular epidemiology and antimicrobial resistance features of *Acinetobacter baumannii* clinical isolates from Pakistan. *Ann. Clin. Microbiol. Antimicrob.* 19, 1–13.
- Khatun, M.N., Farzana, R., Lopes, B.S., Shamsuzzaman, S.M., 2016. Molecular characterization and resistance profile of nosocomial *Acinetobacter baumannii* in intensive care unit of tertiary care hospital in Bangladesh. *Bangladesh Med. Res. Counc. Bull.* 41, 101–107.
- Khoshnood, S., Sadeghifard, N., Mahdian, N., Heidary, M., Mahdian, S., Mohammadi, M., Maleki, A., Haddadi, M.H., 2023. Antimicrobial resistance and biofilm formation capacity among *Acinetobacter baumannii* strains isolated from patients with burns and ventilator-associated pneumonia. *J. Clin. Lab. Anal.* 37, e24814.
- Kumkar, S.N., Kamble, E.E., Chavan, N.S., Dhotre, D.P., Pardesi, K.R., 2022. Diversity of resistant determinants, virulence factors, and mobile genetic elements in *Acinetobacter baumannii* from India: a comprehensive *in silico* genome analysis. *Front. Cell. Infect. Microbiol.* 12, 997897.
- Kyriakidis, I., Vasileiou, E., Pana, Z.D., Tragiannidis, A., 2021. *Acinetobacter baumannii* antibiotic resistance mechanisms. *Pathogens* 10, 373.
- Lam, M.M.C., Wick, R.R., Judd, L.M., Holt, K.E., Wyres, K.L., 2022. Kaptive 2.0: updated capsule and lipopolysaccharide locus typing for the *Klebsiella pneumoniae* species complex. *Microb. Genomics* 8, 800.
- Larsen, M.V., Cosentino, S., Lukjancenko, O., Saputra, D., Rasmussen, S., Hasman, H., Sicheritz-Pontén, T., Aarestrup, F.M., Ussery, D.W., Lund, O., 2014. Benchmarking of methods for genomic taxonomy. *J. Clin. Microbiol.* 52, 1529–1539.
- Leticun, I., Bork, P., 2021. Interactive tree of life (iTOL) v5: an online tool for phylogenetic tree display and annotation. *Nucleic Acids Res.* 49, W293–W296.
- Levy-Blitchtein, S., Roca, I., Plasencia-Rebata, S., Vicente-Taboada, W., Velásquez-Pomar, J., Muñoz, L., Moreno-Morales, J., Pons, M.J., del Valle-Mendoza, J., Vila, J., 2018. Emergence and spread of carbapenem-resistant *Acinetobacter baumannii* international clones II and III in Lima, Peru. *Emerg. Microbes Infect.* 7, 1–9.
- Li, J., Li, Y., Cao, X., Zheng, J., Zhang, Y., Xie, H., Li, C., Liu, C., Shen, H., 2023. Genome-wide identification and oxacillinase OXA distribution characteristics of *Acinetobacter* spp. based on a global database. *Front. Microbiol.* 14, 1174200.
- Lim, S.M.S., Abidin, A.Z., Liew, S.M., Roberts, J.A., Sime, F.B., 2019. The global prevalence of multidrug-resistance among *Acinetobacter baumannii* causing hospital-acquired and ventilator-associated pneumonia and its associated mortality: a systematic review and meta-analysis. *J. Infect.* 79, 593–600.
- Liu, C., Chang, Y., Xu, Y., Luo, Y., Wu, L., Mei, Z., Li, S., Wang, R., Jia, X., 2018. Distribution of virulence-associated genes and antimicrobial susceptibility in clinical *Acinetobacter baumannii* isolates. *Oncotarget* 9, 21663.
- Loehfelm, T.W., Luke, N.R., Campagnari, A.A., 2008. Identification and characterization of an *Acinetobacter baumannii* biofilm-associated protein. *J. Bacteriol.* 190, 1036–1044.
- Mehrotra, T., Konar, D., Pragasam, A.K., Kumar, S., Jana, P., Babele, P., Paul, D., Purohit, A., Tanwar, S., Bakshi, S., 2023. Antimicrobial resistance heterogeneity among multidrug-resistant gram-negative pathogens: phenotypic, genotypic, and proteomic analysis. *Proc. Natl. Acad. Sci.* 120, e2305465120.
- Meier-Kolthoff, J.P., Göker, M., 2019. TYGS is an automated high-throughput platform for state-of-the-art genome-based taxonomy. *Nat. Commun.* 10, 1–10.
- Minh, B.Q., Schmidt, H.A., Chernomor, O., Schrempf, D., Woodhams, M.D., Von Haeseler, A., Lanfear, R., 2020. IQ-TREE 2: new models and efficient methods for phylogenetic inference in the genomic era. *Mol. Biol. Evol.* 37, 1530–1534.
- Morgado, S.M., Fonseca, É.L., Freitas, F.S., Bigli, N.S., Oliveira, P.P.C., Monteiro, P.M., Lima, L.S., Santos, B.P., Sousa, M.A.R., Assumpção, A.O., 2023. Outbreak of high-risk XDR CRAB of international clone 2 (IC2) in Rio Janeiro, Brazil. *J. Glob. Antimicrob. Resist.* 34, 91–98.
- Mussi, M.A., Limansky, A.S., Relling, V., Ravasi, P., Arakaki, A., Actis, L.A., Viale, A.M., 2011. Horizontal gene transfer and assortative recombination within the *Acinetobacter baumannii* clinical population provide genetic diversity at the single *carO* gene, encoding a major outer membrane protein channel. *J. Bacteriol.* 193, 4736–4748.
- Muzahid, N.H., Hussain, M.H., Huët, M.A.L., Dwiyo, J., Su, T.T., Reidpath, D., Mustapha, F., Ayub, Q., Tan, H.S., Rahman, S., 2023. Molecular characterization and comparative genomic analysis of *Acinetobacter baumannii* isolated from the community and the hospital: an epidemiological study in Segamat, Malaysia. *Microb. genomics* 9, 977.
- Nasr, P., 2020. Genetics, epidemiology, and clinical manifestations of multidrug-resistant *Acinetobacter baumannii*. *J. Hosp. Infect.* 104, 4–11.
- Nigro, S.J., Hall, R.M., 2016. Structure and context of *Acinetobacter* transposons carrying the *oxa23* carbapenemase gene. *J. Antimicrob. Chemother.* 71, 1135–1147.
- Odihi, E.E., Oaikhen, A.O., Underwood, A., Houmanou, Y.M.G., Oduyibo, O.O., Fadeyi, A., Aboderin, A.O., Ogunleye, V.O., Argimón, S., Akpunonu, V.N., 2023. High genetic diversity of carbapenem-resistant *Acinetobacter baumannii* isolates recovered in Nigerian hospitals in 2016 to 2020. *MSphere* 8, e00098–23.
- Page, A.J., Cummins, C.A., Hunt, M., Wong, V.K., Reuter, S., Holden, M.T.G., Fookes, M., Falush, D., Keane, J.A., Parkhill, J., 2015. Roary: rapid large-scale prokaryote pan genome analysis. *Bioinformatics* 31, 3691–3693.
- Page, A.J., Taylor, B., Delaney, A.J., Soares, J., Seemann, T., Keane, J.A., Harris, S.R., 2016. SNP-steps: rapid efficient extraction of SNPs from multi-FASTA alignments. *Microb. Genomics* 2, e000056.
- Rahman, A., Styczynski, A., Khaleque, A., Hossain, S.A., Sadique, A., Hossain, A., Jain, M., Tabassum, S.N., Khan, F., Bhuiyan, M.S.S., 2022. Genomic landscape of prominent XDR *Acinetobacter* clonal complexes from Dhaka, Bangladesh. *BMC Genomics* 23, 802.
- Roy, S., Chowdhury, G., Mukhopadhyay, A.K., Dutta, S., Basu, S., 2022. Convergence of biofilm formation and antibiotic resistance in *Acinetobacter baumannii* infection. *Front. Med.* 9, 793615.
- Sánchez-Urtaza, S., Ocampo-Sosa, A., Molins-Bengoetxea, A., El-Kholy, M.A., Hernandez, M., Abad, D., Shawky, S.M., Alkorta, I., Gallego, L., 2023. Molecular characterization of multidrug resistant *Acinetobacter baumannii* clinical isolates from Alexandria, Egypt. *Front. Cell. Infect. Microbiol.* 13.
- Seemann, T., 2014. Prokka: rapid prokaryotic genome annotation. *Bioinformatics* 30, 2068–2069.
- Shadan, A., Pathak, A., Ma, Y., Pathania, R., Singh, R.P., 2023. Deciphering the virulence factors, regulation, and immune response to *Acinetobacter baumannii* infection. *Front. Cell. Infect. Microbiol.* 13, 1053968.
- Shen, M., Luan, G., Wang, Y., Chang, Y., Zhang, C., Yang, J., Deng, S., Ling, B., Jia, X., 2016. Coexistence of blaOXA-23 with armA in quinolone-resistant *Acinetobacter baumannii* from a Chinese university hospital. *Diagn. Microbiol. Infect. Dis.* 84, 230–231.
- Simão, F.A., Waterhouse, R.M., Ioannidis, P., Kriventseva, E.V., Zdobnov, E.M., 2015. BUSCO: assessing genome assembly and annotation completeness with single-copy orthologs. *Bioinformatics* 31, 3210–3212.
- Stepanović, S., Vuković, D., Hola, V., Bonaventura, G.D.I., Djukić, S., Čirković, I., Ruzicka, F., 2007. Quantification of biofilm in microtiter plates: overview of testing conditions and practical recommendations for assessment of biofilm production by staphylococci. *Apmis* 115, 891–899.
- Tacconelli, E., Magrini, N., Kahlmeter, G., Singh, N., 2017. Global priority list of antibiotic-resistant bacteria to guide research, discovery, and development of new antibiotics. *World Heal. Organ.* 27.
- Tada, T., Miyoshi-Akiyama, T., Shimada, K., Ng, T.T.T., Thu, L.T.A., Son, N.T., Ohmagari, N., Kirikae, T., 2015. Dissemination of clonal complex 2 *Acinetobacter baumannii* strains co-producing carbapenemases and 16S rRNA methylase ArmA in Vietnam. *BMC Infect. Dis.* 15, 1–7.
- Tada, T., Uchida, H., Hishinuma, T., Watanabe, S., Tohya, M., Kuwahara-Arai, K., Mya, S., Zan, K.N., Kirikae, T., Tin, H.H., 2020. Molecular epidemiology of multidrug-resistant *Acinetobacter baumannii* isolates from hospitals in Myanmar. *J. Glob. Antimicrob. Resist.* 22, 122–125.
- Tettelin, H., Riley, D., Cattuto, C., Medini, D., 2008. Comparative genomics: the bacterial pan-genome. *Curr. Opin. Microbiol.* 11, 472–477.
- Uddin, B.M.M., Ritu, S., Zubair, A.R., Surovi, E.S., Shamsuzzaman, S.M., 2020. In vitro and in vivo evaluation of antibiotic combination against imipenem resistant

- Acinetobacter Baumannii* strains isolated from Bangladeshi patients. Am. J. Infect. Dis. 8, 83–87.
- Uppalapati, S.R., Sett, A., Pathania, R., 2020. The outer membrane proteins OmpA, CarO, and OprD of *Acinetobacter baumannii* confer a two-pronged defense in facilitating its success as a potent human pathogen. Front. Microbiol. 11, 589234.
- Valenzuela, J.K., Thomas, L., Partridge, S.R., van der Reijden, T., Dijkshoorn, L., Iredell, J., 2007. Horizontal gene transfer in a polyclonal outbreak of carbapenem-resistant *Acinetobacter baumannii*. J. Clin. Microbiol. 45, 453–460.
- Wang, M., Goh, Y.-X., Tai, C., Wang, H., Deng, Z., Ou, H.-Y., 2022. VRprofile2: detection of antibiotic resistance-associated mobilome in bacterial pathogens. Nucleic Acids Res. 50, W768–W773.
- Wareth, G., Linde, J., Nguyen, N.H., Nguyen, T.N.M., Sprague, L.D., Pletz, M.W., Neubauer, H., 2021. WGS-based analysis of carbapenem-resistant *Acinetobacter baumannii* in Vietnam and molecular characterization of antimicrobial determinants and MLST in Southeast Asia. Antibiotics 10, 563.
- Xie, Z., Tang, H., 2017. ISEScan: automated identification of insertion sequence elements in prokaryotic genomes. Bioinformatics 33, 3340–3347.
- Yang, Y., Fu, Y., Lan, P., Xu, Q., Jiang, Y., Chen, Y., Ruan, Z., Ji, S., Hua, X., Yu, Y., 2018. Molecular epidemiology and mechanism of sulbactam resistance in *Acinetobacter baumannii* isolates with diverse genetic backgrounds in China. Antimicrob. Agents Chemother. 62, 10–1128.
- Yu, K., Zeng, W., Xu, Y., Liao, W., Xu, W., Zhou, T., Cao, J., Chen, L., 2021. Bloodstream infections caused by ST2 *Acinetobacter baumannii*: risk factors, antibiotic regimens, and virulence over 6 years period in China. Antimicrob. Resist. Infect. Control 10, 1–9.